Trigonometric Functions & the Unit Circle

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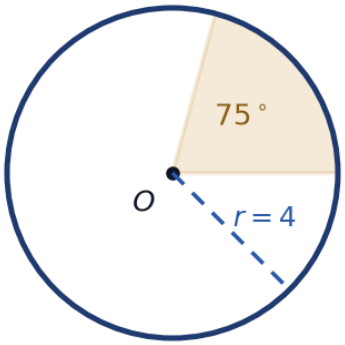
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A terminal ray in standard position passes through the point $P(-8, 15)$. Find the exact values of $\sin\theta$, $\cos\theta$, and $\tan\theta$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given vectors $\mathbf{u} = \langle 4, -3 \rangle$ and $\mathbf{v} = \langle -1, 5 \rangle$, compute: (a) $2\mathbf{u} - 3\mathbf{v}$, (b) the magnitude $||\mathbf{u}||$ and unit vector in the direction of $\mathbf{u}$, and (c) the dot product $\mathbf{u} \cdot \mathbf{v}$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

State the amplitude, period, phase shift, and midline of $f(x) = 3\sin(2x - \pi) + 1$.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A population doubles every 12 years, starting from $P_0 = 500$. When does it reach 4000?

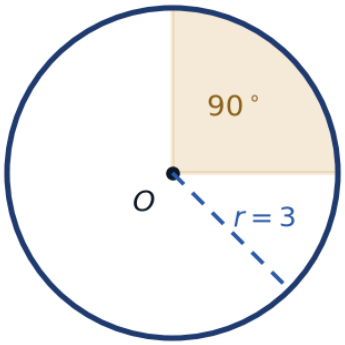
5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Convert the polar equation $r = 6\cos\theta - 8\sin\theta$ to Cartesian rectangular form, and identify the geometric curve, its center, and its radius.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Eliminate the parameter t from the parametric curve $x(t) = 3 + 4\cos t$, $y(t) = -2 + 5\sin t$ for $0 \le t \le 2\pi$. Identify the conic section, its center, and its vertices.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A particle moves along $x(t) = 2t + 1,\ y(t) = t^2 - 3.$ Find the Cartesian equation of the path.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the transformation grid below. If the original point A is mapped to A', what is the vector representing this translation?

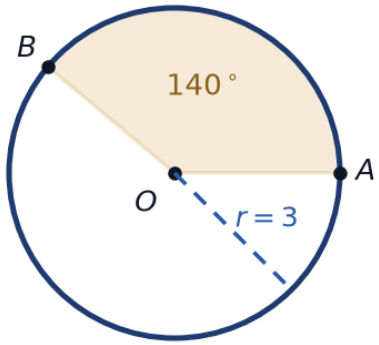
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 3 · AdvancedFind all values of θ in $[0, 2\pi)$ for which the rose curve $r = 4\sin(3\theta)$ passes through the pole (origin), and find the maximum distance of any point on the curve from the origin.

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Unit 3 · Trigonometric and Polar Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedFind the matrix R that rotates vectors by $\theta = \frac{2\pi}{3}$ radians counterclockwise, and use it to find the image of the vector $\vec{w} = \langle 4, -2 \rangle$. Verify that the magnitude of $\vec{w}$ is preserved under the transformation.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 2 · MediumVector $\vec{u} = \langle 4, -3 \rangle$ and vector $\vec{v} = \langle 1, 7 \rangle$. Calculate the magnitude of each vector, determine the dot product $\vec{u} \cdot \vec{v}$, and find the exact angle θ between them in radians.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the trigonometric equation $2\cos^2 x - \sin x - 1 = 0$ for all x in the interval $[0, 2\pi)$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Prove the trigonometric identity $\frac{\sin(2x)}{1 + \cos(2x)} = \tan x$ and state the domain exclusions.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

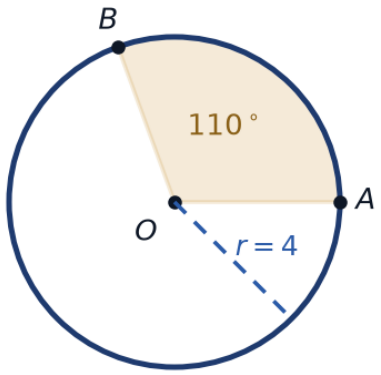
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Level 3 · AdvancedAn invasive plant covers an area modeled by $A(t) = 120(1.15)^t$ square meters, where t is measured in weeks. Write the equation setup to determine when the area reaches 500 square meters, and find the value of t to three decimal places with correct units.

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.03 Rates of Change & Concavity

[AP Precalculus Exam Format & Modeling Framework · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Exponential Functions & Growth Models

[Unit 4 · Parameters, Vectors, and Matrices · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Verify the trigonometric identity: $\frac{\sin(2x)}{\sin x} - \frac{\cos(2x)}{\cos x} = \sec x$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Inverse & Composite Functions

[Unit Circle & Trigonometric Identities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What are the exact coordinates of the terminal point on the unit circle for angle $\theta = \frac{5\pi}{6}$?

a. $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

b. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

c. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

d. $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert an angle of $\frac{7\pi}{4}$ radians into degree measure.

a. 315°

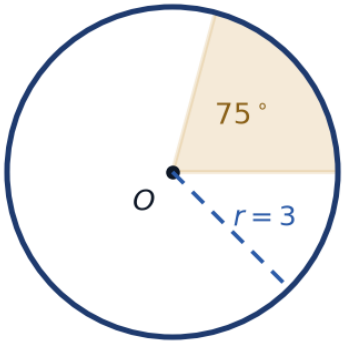
b. 225°

c. 300°

d. 135°

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If the terminal ray of angle α passes through the point $(-3, -4)$, what is $\sin\alpha$?



a. $-\frac{4}{5}$

b. $-\frac{3}{5}$

c. $\frac{4}{3}$

d. $\frac{3}{5}$

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the exact value of $\sec\left(-\frac{\pi}{4}\right)$?

a. $\sqrt{2}$

b. $-\sqrt{2}$

c. $\frac{\sqrt{2}}{2}$

d. 2

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the amplitude and period of the sinusoidal function $f(x) = -4\cos(3x - \pi) + 2$.

a. Amplitude $= 4$, Period $= \frac{2\pi}{3}$

b. Amplitude $= -4$, Period $= 3$

c. Amplitude $= 4$, Period $= 3\pi$

d. Amplitude $= 2$, Period $= \frac{2\pi}{3}$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A Ferris wheel of radius 25 m completes one rotation every 60 seconds. Passengers board at the bottom, 2 m above the ground. Which cosine equation models height $h(t)$ in meters?

a. $h(t) = -25\cos\left(\frac{\pi}{30}t\right) + 27$

b. $h(t) = 25\cos\left(\frac{\pi}{30}t\right) + 27$

c. $h(t) = -25\cos\left(\frac{\pi}{60}t\right) + 2$

d. $h(t) = -27\cos\left(\frac{\pi}{30}t\right) + 25$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the phase shift of $g(x) = 2\sin\left(4x + \frac{\pi}{2}\right)$?

a. $\frac{\pi}{8}$ units to the left

b. $\frac{\pi}{8}$ units to the right

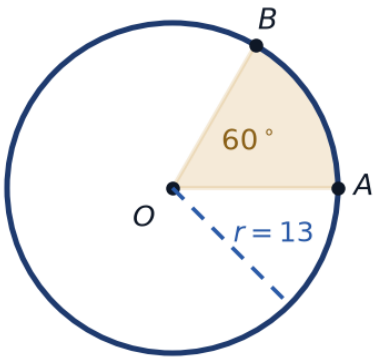
c. $\frac{\pi}{2}$ units to the left

d. $\frac{\pi}{4}$ units to the left

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Simplify the trigonometric expression $\frac{\tan x}{\sec x}$.



a. $\sin x$

b. $\cos x$

c. $\csc x$

d. $\cot x$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which expression is equivalent to $(1 - \cos^2\theta)(1 + \cot^2\theta)$ for all defined θ?

a. 1

b. $\sin^2\theta$

c. $\tan^2\theta$

d. $\cos^2\theta$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A circle has radius 5. Find the arc length subtended by a central angle of $\frac{1\pi}{2}$ radians.

a. $5/4\pi$

b. $5/2\pi$

c. $7/2\pi$

d. 5π

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Eliminate the parameter: $x = 3t$, $y = 3t$.

a. $y = 1x$

b. $y = 1x + 1$

c. $y = 2x$

d. $y = -1x$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Let $f(x) = 3x + 2$ and $g(x) = 1/3x - 2/3$. Are f and g inverse functions?

a. No, f and g are not inverses

b. Yes, f and g are inverses

c. Cannot be determined

d. Only if $x = 0$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

End behaviour of $f(x) = -2x^5$ as $x \to +\infty$:



a. $+\infty$

b. $-\infty$

c. 0

d. 2

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the determinant of the 2×2 coefficient matrix $A = (4, 2; 3, 4)$.

a. 22

b. 10

c. 13

d. 6

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Which transformation shifts $f(x)$ right by 3?

a. $f(x+3)$

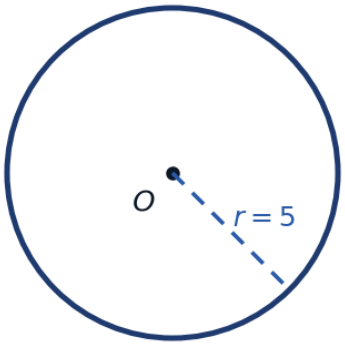
b. $f(x-3)$

c. $f(x)+3$

d. $f(x)-3$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.



a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

c. 1

d. $\frac{\sqrt{2}}{2}$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the vector $\mathbf{v}$ directed from the point $P = (2, 5)$ to the point $Q = (7, 1)$. What is the component form of $\mathbf{v}$?

a. $\langle -5, -4 \rangle$

b. $\langle 5, 4 \rangle$

c. $\langle 5, -4 \rangle$

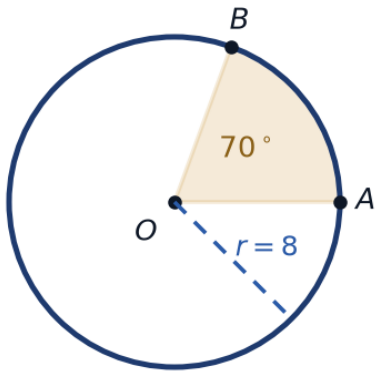
d. $\langle -5, 4 \rangle$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.02 Logarithmic Functions & Properties

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?



a. $\cos\theta$

b. $\sin\theta$

c. $\tan\theta$

d. 1

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Inverse & Composite Functions

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let θ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

a. $-\frac{3}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$

d. $\frac{4}{5}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.03 Parametric Equations & Modeling

Consider a point P on the unit circle in standard position. If the angle θ in standard position has a terminal side passing through P such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. -1

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. 1

d. 5/3

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.02 Rational Functions & Asymptotes

On the unit circle, a point $P(x, y)$ corresponds to an angle θ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

a. $\frac{1}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$

d. 1

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.03 Rates of Change & Concavity

What is the maximum value of the function $f(x) = \sqrt{3}\sin(2x) - \cos(2x) - 2\sin x + 2\sqrt{3}\cos x$ for all real numbers x?

a. 3

b. 2

c. 6

d. 1

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